JAVA PROJECT REPORT ON

PHARMACY MEDICINE STOCK AND BILLING MANAGEMENT SYSTEM

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Academic Year: 2025 – 2026

APRIL 2026

Semester: II

DECLARATION

I hereby declare that the project entitled "Pharmacy Medicine Stock and Billing Management System" submitted in partial fulfillment of the requirements for the course [JAVA PROGRAMMING] is a record of my original work carried out under the guidance of [DHILEEBAN A], Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

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CERTIFICATE

This is to certify that the project entitled **"Pharmacy Medicine Stock and Billing Management System"** is a bona fide work carried out by **[KAVIYA K] [I-B.Sc IT 'A']**, [B.Sc Information Technology], KG College of Arts & Science, in partial fulfillment of the requirements for the course [JAVA PROGRAMMING].

This project has been completed under my supervision and guidance during the academic period April 2026.

Place:

Date:

Submitted for the Viva-Voce Examination held on

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ABSTRACT

The Pharmacy Medicine Stock and Billing Management System is a Java-based application designed to automate and streamline the management of medicine inventory and customer billing operations in pharmacies and medical stores. In many small and medium pharmacies, stock management and billing are performed manually, which can lead to inventory errors, billing mistakes, expired stock oversight, and inefficient record-keeping. This project aims to solve these problems by providing a comprehensive automated system.

The system allows pharmacy staff to add and manage medicine stock, track expiry dates, search for medicines by name or category, process customer purchases, and generate accurate bills including GST. The application is developed using Core Java programming concepts, including classes, objects, methods, conditional statements, loops, and exception handling. The project is implemented as a console-based application or using Java Swing for a graphical user interface.

The main objective of the system is to improve billing efficiency, maintain accurate stock levels, alert staff to low stock and expiry situations, and reduce human errors. This project demonstrates the practical implementation of object-oriented programming concepts in Java and provides a strong foundation for developing more advanced pharmacy management systems in the future.

1. INTRODUCTION

Pharmacies play a critical role in the healthcare system by supplying medicines and health products to patients. Efficient management of medicine stock and accurate billing are essential for the smooth operation of any pharmacy. Traditionally, many small pharmacies rely on manual methods to track medicines, manage inventory, and generate bills. This manual approach is prone to errors, time-consuming, and can lead to serious issues such as dispensing expired medicines or running out of critical stock.

With the advancement of computer technology, automated management systems have become indispensable for improving the efficiency and accuracy of pharmacy operations. A computerized pharmacy system helps staff manage stock in real time, generate bills instantly, track expiry dates, and maintain records of all transactions.

The Pharmacy Medicine Stock and Billing Management System is a Java-based application designed to automate all core pharmacy operations. The system enables staff to add new medicines, update stock, search for medicines, process customer bills with GST calculation, and generate receipts. It provides alerts for low stock and medicines nearing expiry.

This project is developed using Core Java programming concepts such as classes, objects, methods, conditional statements, loops, arrays, and exception handling. The primary goal is to reduce manual effort, prevent billing errors, and ensure better inventory control in pharmacy environments.

1.1 Objective

The main objective of the Pharmacy Medicine Stock and Billing Management System is to develop a reliable and efficient application using Core Java that automates medicine stock management and billing operations in a pharmacy.

The specific objectives of this project are:

- To design and develop a Java-based pharmacy management system.
- To automate the management of medicine stock including additions, updates, and deletions.
- To track medicine expiry dates and generate alerts for expired or near-expiry stock.
- To provide low-stock alerts to prevent out-of-stock situations.
- To automate customer billing including GST calculation and invoice generation.
- To enable searching of medicines by name and category.
- To demonstrate the implementation of Core Java OOP concepts in a real-world application.

1.2 Overview

The Pharmacy Medicine Stock and Billing Management System is a menu-driven application developed using the Java programming language. The system provides a digital platform where pharmacy staff can manage all aspects of medicine inventory and customer billing from a single interface.

The system displays a medicine catalog with prices, stock quantities, and expiry information. Staff can add new medicines to the inventory, update existing stock, remove discontinued medicines, and search for medicines by name or category. During billing, the system retrieves medicine prices from the inventory, calculates the total amount based on quantity, applies applicable GST, and generates a structured invoice.

The system helps pharmacies eliminate manual stock registers and paper-based billing, thereby reducing errors and improving customer service speed. It can be developed as a console-based program or extended to a Java Swing graphical user interface.

1.3 Features

The Pharmacy Medicine Stock and Billing Management System includes the following main features:

- Medicine Stock Management – Add, update, and delete medicine records including name, category, price, quantity, and expiry date.
- Stock Search – Search for medicines by name or category for quick retrieval.
- Expiry Date Tracking – Automatically identifies and alerts staff about expired or near-expiry medicines.
- Low Stock Alerts – Notifies staff when medicine stock falls below a defined threshold.
- Billing Module – Processes customer purchases by selecting medicines and quantities, with automatic price retrieval.
- GST Calculation – Automatically computes GST and adds it to the total bill.
- Invoice Generation – Generates a detailed receipt showing items, quantities, prices, subtotal, tax, and final amount.
- User-Friendly Interface – Provides a simple menu-driven interface for easy operation by pharmacy staff.
- Stock Quantity Update – Automatically reduces stock quantities after each sale.

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying current issues in manual pharmacy operations and proposing a computerized solution that improves efficiency, accuracy, and safety.

2.1 Problem Definition

In many small pharmacies and medical stores, medicine stock management and billing are performed manually using paper registers and basic calculators. This traditional method creates numerous operational challenges.

Some common problems include:

- Manual calculation errors in billing leading to incorrect charges.
- Difficulty tracking large inventories of medicines across multiple categories.
- Risk of dispensing expired medicines due to lack of automated expiry tracking.
- Stock-outs caused by poor visibility of inventory levels.
- Slow billing process leading to longer customer waiting times.
- No systematic records of sales transactions for audit or analysis.
- Increased workload on pharmacy staff during peak hours.

Therefore, there is a need for an automated pharmacy management system that can manage stock accurately, track expiry dates, alert staff to critical situations, and generate bills instantly.

2.2 Project Overview

The Pharmacy Medicine Stock and Billing Management System is a Java-based application developed to address all the above problems. It provides an integrated solution for medicine inventory management and customer billing operations.

The system provides the following functionalities:

- Adding and maintaining a comprehensive medicine inventory with details such as name, category, manufacturer, price, quantity, and expiry date.
- Searching for medicines by name or category for quick access.
- Monitoring stock levels and generating low-stock alerts.
- Tracking expiry dates and flagging expired or soon-to-expire medicines.
- Processing customer bills by selecting medicines and quantities.
- Automatically calculating subtotal, GST, and final payable amount.
- Generating structured invoices for customers.
- Updating stock quantities automatically after each sale.

2.3 Scopes and Limitations

Scope

The scope of the Pharmacy Medicine Stock and Billing Management System includes:

- Complete medicine stock management including CRUD operations.
- Customer billing with automatic price lookup and GST calculation.
- Expiry date monitoring and alert generation.
- Low-stock threshold alerts.
- Sales transaction record keeping.
- Applicable in pharmacies, medical stores, hospital dispensaries, and health clinics.

Limitations

- The current system does not include a database connection; data is managed in memory during program execution.
- The system does not support multi-user concurrent access in its basic form.
- It does not include supplier management or purchase order functionality in the base version.
- Advanced reporting and analytics are not included in the initial implementation.

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules and the workflow of the application is defined. Proper system design ensures organized development and easier maintenance.

The Pharmacy Medicine Stock and Billing Management System is designed as a menu-driven application where users can perform stock operations and generate customer bills through a series of structured options.

3.1 Modules

The main modules of the system are explained below:

1. Medicine Stock Management Module

This module manages the complete medicine inventory. It allows pharmacy staff to:

- Add new medicines with details: name, category, manufacturer, price, quantity, expiry date.
- Update existing medicine records such as price or quantity.
- Delete discontinued medicines from the system.
- View the complete list of available medicines.

Example Medicine Record:

```
Medicine ID: M001 | Name: Paracetamol | Category: Analgesic | Price: ₹2.50 | Qty: 500 | Expiry: 12/2026
```

2. Stock Search Module

This module enables quick retrieval of medicine information:

- Search by medicine name for exact or partial matches.
- Search by category to list all medicines in a group (e.g., Antibiotics, Analgesics).
- Display price, availability, and expiry details of the searched medicine.

3. Expiry Date Tracking Module

This module automatically monitors the expiry status of all medicines:

- Identifies medicines that have already expired.
- Flags medicines expiring within the next 30 days.
- Displays an alert list for staff action.
- Prevents billing of expired medicines.

4. Low Stock Alert Module

This module monitors inventory levels and generates alerts:

- Checks stock quantity against a defined minimum threshold.
- Displays a list of medicines that need to be reordered.
- Helps prevent stock-out situations for critical medicines.

5. Billing Module

This module processes customer purchases and generates bills:

- Allows staff to select medicines from the inventory by name or ID.
- Accepts quantity input for each selected medicine.
- Retrieves unit prices from the inventory automatically.
- Calculates item totals, subtotal, GST (e.g., 12%), and final payable amount.
- Reduces stock quantity automatically after billing.

Example Bill Calculation:

```
Item Total = Price × Quantity
GST = 12% of Subtotal
Final Bill = Subtotal + GST
```

6. Invoice Generation Module

This module generates a detailed printed or displayed invoice:

- Invoice header with pharmacy name and date.
- List of all purchased medicines with quantity and price.
- Subtotal, GST amount, and total payable amount.
- Thank you message and return policy (if applicable).

7. Exit Module

This module allows the user to exit the application safely after completing all operations. When the exit option is selected, the system terminates gracefully.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Pharmacy Medicine Stock and Billing Management System, a structured development approach was followed to ensure that the system meets all requirements and functions efficiently.

4.1 Problem Analysis

The main problems identified in manual pharmacy operations:

- Manual stock tracking leads to inaccuracies and data loss.
- Billing errors occur due to manual price lookups and calculations.
- Expired medicines may be accidentally dispensed without a tracking system.
- Stock-out situations arise from lack of real-time inventory visibility.
- Transaction records are difficult to maintain and retrieve manually.

4.2 System Design

The system is divided into the following modules:

- Medicine Stock Management Module
- Stock Search Module
- Expiry Date Tracking Module
- Low Stock Alert Module
- Billing Module
- Invoice Generation Module
- Exit Module

4.3 Program Development

The program is implemented using the Java programming language with the following steps:

- Creation of a Medicine class to encapsulate all medicine attributes.
- Implementation of an ArrayList to store medicine records dynamically.
- Development of menu-driven logic using switch-case statements.
- Writing methods for stock addition, update, deletion, and search operations.
- Implementation of billing logic with automatic price retrieval and GST calculation.
- Handling user input using the Scanner class with appropriate exception handling.
- Displaying formatted output for invoices and stock reports.

4.4 Testing and Validation

Different test cases were used to verify the correctness and robustness of the system:

Test Case	Expected Result
Add a new medicine	Medicine added to inventory with correct details
Search medicine by name	Correct medicine details displayed
Bill an available medicine	Correct total and GST calculated, stock reduced
Attempt to bill expired medicine	Alert displayed, billing blocked
Enter invalid quantity (negative)	Error handled gracefully
View low stock alert	Medicines below threshold listed correctly
Update medicine price	New price reflected in all subsequent billing
Delete a medicine	Medicine removed from inventory

4.5 Result and Documentation

After successful testing, the system was found to correctly perform all required operations. Stock management functions work accurately, billing calculations including GST are correct, and alerts for expired and low-stock medicines are generated as expected. Program execution screenshots and invoice outputs are documented in Chapter 6.

4.6 Algorithm

Algorithm for the Pharmacy Medicine Stock and Billing Management System:

1. Start the program.
2. Display the main menu with options: Manage Stock, Search Medicine, Billing, View Alerts, Exit.
3. If 'Manage Stock' is selected: display sub-menu for Add/Update/Delete/View medicines.
4. If 'Add Medicine': accept input for medicine name, category, price, quantity, expiry date and store in inventory.
5. If 'Billing' is selected: accept customer name and start item selection.
6. For each item: search inventory by name, retrieve price, accept quantity, calculate item total.
7. After all items are added: calculate subtotal, apply 12% GST, compute final bill.
8. Check that selected medicines are not expired and have sufficient stock.
9. Generate and display the formatted invoice.
10. Update inventory quantities after billing.
11. If 'View Alerts': scan inventory for expired medicines and medicines below minimum stock threshold.
12. If 'Exit': terminate the program.
13. End.

4.7 Flowchart

The flowchart below illustrates the program flow of the Pharmacy Medicine Stock and Billing Management System:

```
Start
↓
Display Main Menu
↓
User Selection → [Manage Stock / Search / Billing / Alerts / Exit]
↓ (Billing path)
Select Medicine & Quantity → Check Expiry & Stock
↓
Calculate Subtotal + GST → Generate Invoice → Update Stock
↓
End
```

Students can draw a detailed flowchart using draw.io, Lucidchart, or MS Visio.

5. SOURCE CODE

The following are the key parts of the Java source code for the Pharmacy Medicine Stock and Billing Management System:

Medicine Class

```
class Medicine {
    String id, name, category, manufacturer;
    double price;
    int quantity;
    String expiryDate;

    Medicine(String id, String name, String category,
             String manufacturer, double price, int quantity, String expiryDate) {
        this.id = id;
        this.name = name;
        this.category = category;
        this.manufacturer = manufacturer;
        this.price = price;
        this.quantity = quantity;
        this.expiryDate = expiryDate;
    }
}
```

Billing Calculation

```
double subtotal = 0;
for (BillItem item : billItems) {
    subtotal += item.price * item.quantity;
}
double gst = subtotal * 0.12;
double finalBill = subtotal + gst;
```

Invoice Generation

```
System.out.println("========== PHARMACY BILL ==========");
System.out.println("Medicine Name   Qty   Price   Total");
for (BillItem item : billItems) {
    System.out.printf("%-16s %-5d %-7.2f %.2f%n",
        item.name, item.quantity, item.price, item.price * item.quantity);
}
System.out.printf("Subtotal: %.2f%n", subtotal);
System.out.printf("GST (12%%): %.2f%n", gst);
System.out.printf("Total Payable: %.2f%n", finalBill);
```

6. INPUT AND OUTPUT SCREENS

Include screenshots of the following screens after running the program:

- Figure 1: Main Menu Screen
- Figure 2: Add Medicine Screen
- Figure 3: Medicine Inventory List
- Figure 4: Search Medicine Result
- Figure 5: Billing Screen – Item Selection
- Figure 6: Invoice / Bill Output
- Figure 7: Low Stock Alert Screen
- Figure 8: Expiry Alert Screen

Each screenshot should be labeled with its figure number and a brief description. Students should capture actual output from their console or Java Swing window.

Example Invoice Output:

```
====================================
       MEDCARE PHARMACY
  Medicine Billing System
  Date: 09/04/2026
====================================
Medicine         Qty   Rate    Total
------------------------------------
Paracetamol 500mg  10   2.50   25.00
Amoxicillin 250mg   6  15.00   90.00
Vitamin C 500mg    12   5.00   60.00
------------------------------------
Subtotal:              175.00
GST (12%):              21.00
Total Payable:         196.00
====================================
Thank you! Get well soon.
====================================
```

7. DISCUSSION AND CONCLUSION

7.1 Discussion

The Pharmacy Medicine Stock and Billing Management System was developed to automate and streamline the day-to-day operations of a pharmacy. The system successfully integrates medicine inventory management, billing, expiry tracking, and stock alert functionalities into a single application.

During the development process, Core Java programming concepts such as classes, objects, arrays, ArrayList, loops, conditional statements, exception handling, and methods were applied effectively. The Medicine class encapsulates all medicine attributes and supports the object-oriented design of the system.

The billing module successfully retrieves medicine prices from the inventory, calculates totals for multiple items, applies 12% GST, and generates formatted invoices. The expiry tracking module scans the inventory and identifies expired and near-expiry medicines, helping pharmacies avoid dispensing unsafe products.

The system was tested with various scenarios including normal billing, boundary cases such as zero quantity input, expired medicine billing attempts, and invalid input handling. The results confirmed that the system handles all scenarios correctly and produces accurate outputs.

7.2 Conclusion

The Pharmacy Medicine Stock and Billing Management System successfully demonstrates the practical application of Core Java programming concepts in developing a real-world healthcare business application. The system automates medicine stock management, prevents billing of expired medicines, alerts staff to critical stock levels, and generates accurate invoices with GST.

This project significantly reduces manual errors, speeds up the billing process, and improves overall pharmacy management efficiency. It provides pharmacy staff with a reliable tool to maintain accurate inventory and serve customers faster.

The system can be further enhanced by adding database connectivity using JDBC for persistent data storage, a Java Swing graphical user interface, supplier and purchase order management, sales report generation, barcode scanning integration, and online payment support. This project provides a strong foundation for developing a full-featured pharmacy management software.

8. REFERENCES

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